

Concentration for Random Product Formulas

Course Paper — Theory and Numerical Experiments

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Abstract

This part reconstructs the theory of *Concentration for random product formulas* (Chen, Huang, Kueng, Tropp), which analyzes the qDRIFT randomized product formula for Hamiltonian simulation. The paper asks whether qDRIFT's gate count — famously free of explicit dependence on the number L of Hamiltonian terms — comes from *averaging* many random circuits or is already present in *a single* one. The answer is the latter: one realization typically approximates the ideal evolution with high probability, using $\tilde{O}(n\lambda^2 t^2/\epsilon^2)$ gates for a worst-case (diamond-norm) guarantee and $\tilde{O}(\lambda^2 t^2/\epsilon^2)$ for a fixed input, where n is the qubit number and $\lambda = \sum_j \|h_j\|$. The proof decomposes the error into a deterministic bias and a random fluctuation, controlling the latter by a matrix martingale (worst case) or a dimension-free vector martingale (fixed input). Code and figures are available at JinqiSMS/quantum_computing.git.

1 Setup and the Central Question

For $H = \sum_{j=1}^L h_j$ on n qubits, digital simulation approximates $U = e^{-itH}$ by a product $V_N \cdots V_1$ of simple gates, with cost measured by N . The first-order Lie-Trotter formula $e^{-itH} \approx (e^{-i(t/r)h_L} \cdots e^{-i(t/r)h_1})^r$ uses $N = rL$ gates: each layer must cycle through all L terms, an explicit L -dependence that is costly when L is large.

qDRIFT (Campbell) avoids this. With $\lambda = \sum_j \|h_j\|$ and $p_j = \|h_j\|/\lambda$, each step samples one index $j \sim p_j$ and applies

$$V_k = \exp\left(-i \frac{t}{N} X_k\right), \quad X_k = \frac{\lambda}{\|h_j\|} h_j \text{ w.p. } p_j, \quad \text{so} \quad \mathbb{E}[X_k] = \sum_j p_j \frac{\lambda}{\|h_j\|} h_j = H. \quad (1)$$

This unbiasedness gives $\mathbb{E}[V_k] \approx e^{-i(t/N)H} = U^{1/N}$, hence by independence $\mathbb{E}[V_N \cdots V_1] = (\mathbb{E}V)^N \approx U$. Campbell proved the *averaged channel* $\bar{\mathcal{V}}(\rho) = \mathbb{E}[V_N \cdots V_1 \rho V_1^\dagger \cdots V_N^\dagger]$ is ϵ -close to $\mathcal{U}(\rho) = U \rho U^\dagger$ in diamond distance once $N = O(\lambda^2 t^2/\epsilon)$, with *no* explicit L .

That bound concerns the average over circuits, but a user runs the box once and gets a single unitary $V_N \cdots V_1$. The paper asks: *does one random circuit already work?* This separates *random sampling* from *averaging*. The answer: sampling alone removes the L -dependence; averaging only improves the accuracy exponent from $1/\epsilon^2$ to $1/\epsilon$.

Throughout, $d = 2^n$, the generator $(t/N)X_k$ is Hermitian with $\|(t/N)X_k\| = t\lambda/N$ (since $\|X_k\| = \lambda$), and $\mathcal{V}^{(N)}(\rho) = V_N \cdots V_1 \rho V_1^\dagger \cdots V_N^\dagger$ is the single-realization channel. Two error

notions arise. For *all* inputs and observables (worst case), the relevant quantity for unitary channels is the diamond distance,

$$\frac{1}{2}\|\mathcal{U} - \mathcal{V}\|_\diamond = \max_{|\psi\rangle} \frac{1}{2}\|U|\psi\rangle\langle\psi|U^\dagger - V|\psi\rangle\langle\psi|V^\dagger\|_1 = \max_{|\psi\rangle} \max_{\|O\|\leq 1} |\langle O \rangle_{U|\psi\rangle} - \langle O \rangle_{V|\psi\rangle}|; \quad (2)$$

for a single *fixed* input ρ , it is the trace distance $\frac{1}{2}\|U\rho U^\dagger - V\rho V^\dagger\|_1$.

2 Main Results

Theorem 1 (Worst case, diamond norm). *Draw $V^{(N)} = V_N \cdots V_1$ from (1) with $N \geq \Omega((n + \log(1/\delta))\lambda^2 t^2 / \epsilon^2)$. Then with probability $\geq 1 - \delta$, $\max_{|\psi\rangle} \max_{\|O\|\leq 1} |\langle O \rangle_{V^{(N)}|\psi\rangle} - \langle O \rangle_{U|\psi\rangle}| < \epsilon$. Moreover $\mathbb{E}[\frac{1}{2}\|\mathcal{U} - \mathcal{V}^{(N)}\|_\diamond] \lesssim \sqrt{n\lambda^2 t^2 / N}$.*

Theorem 2 (Fixed input, trace norm). *Fix any state $|\psi\rangle$ and draw $V^{(N)}$ with $N \geq \Omega(\lambda^2 t^2 \log(1/\delta) / \epsilon^2)$. Then with probability $\geq 1 - \delta$, $V^{(N)}|\psi\rangle$ is ϵ -close to $U|\psi\rangle$ in trace distance, and $\max_\rho \mathbb{E}[\frac{1}{2}\|\mathcal{U}(\rho) - \mathcal{V}^{(N)}(\rho)\|_1] \lesssim \sqrt{\lambda^2 t^2 / N}$.*

Neither bound contains L , so qDRIFT's L -independence already holds for one circuit; by the probabilistic method Theorem 1 even proves such L -free formulas *exist*. The price for a single circuit is accuracy $1/\epsilon^2$ (vs. $1/\epsilon$ for the average), and the worst case carries a factor $n = \log_2 d$. Strikingly, fixing the input removes n *without any knowledge of the state* — only the assumption that $|\psi\rangle$ is chosen independently of the gates; deterministic methods usually need spectral structure (e.g. a low-energy subspace) for such a gain. Writing $\tilde{O}$ for log-suppressed factors, the hierarchy is

$$N_{\text{avg}} = O\left(\frac{\lambda^2 t^2}{\epsilon}\right) \prec N_{\text{fixed}} = \tilde{O}\left(\frac{\lambda^2 t^2}{\epsilon^2}\right) \prec N_{\text{worst}} = \tilde{O}\left(\frac{n\lambda^2 t^2}{\epsilon^2}\right).$$

The $1/\epsilon$ vs. $1/\epsilon^2$ gap is the “quadratic improvement from mixing,” an instance of Jensen’s inequality for the convex distance functional.

3 Proof Strategy

All regimes share one skeleton: reduce the diamond distance to an operator (or vector) norm, split the error as

$$\|V_N \cdots V_1 - U\| \leq \underbrace{\|(\mathbb{E}V)^N - U\|}_{\text{deterministic bias}} + \underbrace{\|V_N \cdots V_1 - (\mathbb{E}V)^N\|}_{\text{random fluctuation}}, \quad (3)$$

bound the bias by matrix-exponential estimates plus telescoping, and bound the fluctuation by a martingale inequality. Geometrically $V_N \cdots V_1$ is a random walk on $U(d)$: the bias is the offset of its mean from U , the fluctuation its spread about the mean.

Step 1: diamond $\rightarrow$ operator norm. For unitary channels no ancilla is needed, so the diamond distance collapses to a norm.

Lemma 1. *For unitary $\mathcal{U}, \mathcal{V}$ and any pure $|\psi\rangle$, $\frac{1}{2}\|\mathcal{U}(|\psi\rangle\langle\psi|) - \mathcal{V}(|\psi\rangle\langle\psi|)\|_1 \leq \|(U - V)|\psi\rangle\|_{\ell_2}$, hence $\frac{1}{2}\|\mathcal{U} - \mathcal{V}\|_\diamond \leq \|U - V\|$; for the averaged channel, $\frac{1}{2}\|\mathcal{U} - \mathbb{E}[\mathcal{V}]\|_\diamond \leq 2\|U - \mathbb{E}V\|$.*

Proof idea. Fix $|\psi\rangle$ and set $|u\rangle = U|\psi\rangle$, $|v\rangle = V|\psi\rangle$, so $|\langle u, v \rangle| \leq 1$. The Fuchs–van de Graaf relation converts the output trace distance into a fidelity:

$$\frac{1}{2}\| |u\rangle\langle u| - |v\rangle\langle v| \|_1 = \sqrt{1 - |\langle u, v \rangle|^2} = \sqrt{(1 + |\langle u, v \rangle|)(1 - |\langle u, v \rangle|)} \leq \sqrt{2(1 - \operatorname{Re}\langle u, v \rangle)} = \| |u\rangle - |v\rangle \|_{\ell_2}.$$

Maximizing over pure inputs, and using that stabilization (an ancilla) is unnecessary for the diamond distance of unitary channels, gives the first two claims. For random V with $E = \mathbb{E}V$, expand $\mathbb{E}[V|\psi\rangle\langle\psi|V^\dagger] = E|\psi\rangle\langle\psi|E^\dagger + \mathbb{E}[(V - E)|\psi\rangle\langle\psi|(V - E)^\dagger]$ (cross terms vanish); bounding the bias term and the positive-semidefinite variance term separately, each by $\|(U - E)|\psi\rangle\|_{\ell_2}$, yields the factor 2 (due to Allen–Zhao, correcting an earlier claim that omitted it). $\square$

Step 2: deterministic bias. Using $\|e^{iX} - I\| \leq \|X\|$ and $\|e^{iX} - iX - I\| \leq \frac{1}{2}\|X\|^2$ for Hermitian X , with $\|(t/N)X_k\| = t\lambda/N$ and $\mathbb{E}[(t/N)X_k] = (t/N)H$:

$$\|V - \mathbb{E}V\| \leq \frac{2t\lambda}{N}, \quad \|\mathbb{E}V - U^{1/N}\| \leq \left(\frac{t\lambda}{N}\right)^2. \quad (4)$$

The first follows from $\|V - \mathbb{E}V\| \leq \|e^{-iX} - I\| + \mathbb{E}\|I - e^{-iX}\| \leq 2\|X\|$ (with $X = (t/N)X_k$); the second from $\|\mathbb{E}V - U^{1/N}\| \leq \frac{1}{2}\mathbb{E}\|X\|^2 + \frac{1}{2}\|\mathbb{E}X\|^2 \leq \mathbb{E}\|X\|^2$ via the first-order estimate and Jensen. The single-step bias accumulates by a telescoping bound $\|(\mathbb{E}V)^N - (U^{1/N})^N\| \leq N\|\mathbb{E}V - U^{1/N}\|$ (iterate $\|A_1A_2 - B_1B_2\| \leq \|A_1 - B_1\| + \|A_2 - B_2\|$ for norm- ≤ 1 factors), giving the key estimate

$$\|U - \mathbb{E}[V_N \cdots V_1]\| = \|(U^{1/N})^N - (\mathbb{E}V)^N\| \leq \frac{t^2\lambda^2}{N}. \quad (5)$$

This is exactly the averaged-channel bias, $\leq \epsilon/2$ once $N \geq 2t^2\lambda^2/\epsilon$ — already L -free with the favorable $1/\epsilon$ rate. Geometrically, the bias is how far the random walk’s *mean* $(\mathbb{E}V)^N$ sits from the target U ; the next step controls the walk’s *spread* about that mean, and the averaged channel pays only the bias — the root of its quadratic accuracy advantage.

Step 3: fluctuation via a matrix martingale. Interpolate between the two ends of (3) by

$$B_k = (\mathbb{E}V)^{N-k}V_k \cdots V_1, \quad B_0 = (\mathbb{E}V)^N, \quad B_N = V_N \cdots V_1. \quad (6)$$

Adjacent iterates differ only at slot k , where $\mathbb{E}V$ is replaced by a fresh V_k . The process is causal and obeys $\mathbb{E}[B_k | V_{k-1} \cdots V_1] = (\mathbb{E}V)^{N-k}\mathbb{E}[V_k]V_{k-1} \cdots V_1 = B_{k-1}$, so it is a martingale. Its increments satisfy $\|C_k\| = \|(\mathbb{E}V)^{N-k}(V_k - \mathbb{E}V)V_{k-1} \cdots V_1\| \leq \|V_k - \mathbb{E}V\| \leq 2t\lambda/N =: R$ and predictable variation $\|\sum_k \mathbb{E}_{k-1}C_k C_k^\dagger\| \leq NR^2 =: v$. Matrix Freedman, $\Pr[\|B_N - B_0\| \geq \tau] \leq 2d \exp(-\frac{\tau^2/2}{v+R\tau/3})$, then gives:

Proposition 1. For $\epsilon \in [0, 4t\lambda]$, $\Pr[\|V_N \cdots V_1 - \mathbb{E}[V_N \cdots V_1]\| \geq \epsilon/2] \leq 2d \exp(-N\epsilon^2/44t^2\lambda^2)$. Hence $N \geq (44t^2\lambda^2/\epsilon^2) \log(2d/\delta)$ forces the deviation $\leq \epsilon/2$ w.p. $\geq 1 - \delta$.

The error is sub-Gaussian with variance $\sim N(t\lambda/N)^2$, a random walk of N steps of size $t\lambda/N$. The prefactor $2d$ is intrinsic to a *matrix*-valued martingale; under the log it becomes $\log(2d) = O(n)$ — the sole source of the system-size factor. Combining (5) ($\leq \epsilon/2$ for $N \geq 2t^2\lambda^2/\epsilon$) with Proposition 1 ($\log(2d/\delta) = O(n + \log \frac{1}{\delta})$) proves Theorem 1; integrating the tail gives the expected bound.

Step 4: fixed input via a vector martingale. For a fixed $|\psi\rangle$ one compares the output vectors $U|\psi\rangle$ and $V_N \cdots V_1|\psi\rangle$, so (6) becomes the vector martingale $|\psi_k\rangle = V_k \cdots V_1|\psi\rangle$. Vector concentration is *dimension-free*; the tool is uniform smoothness of the ℓ_2 norm: if $\mathbb{E}[y | x] = 0$ then $(\mathbb{E}\|x + y\|_{\ell_2}^q)^{2/q} \leq (\mathbb{E}\|x\|_{\ell_2}^q)^{2/q} + (q-1)(\mathbb{E}\|y\|_{\ell_2}^q)^{2/q}$ for $q \geq 2$. Applied recursively across the increments (whose two pieces are conditionally orthogonal) with $\mathbb{E}\|V_k - \mathbb{E}V_k\|^q \leq (2t\lambda/N)^q$, it yields $(\mathbb{E}\|(V_N \cdots V_1 - \mathbb{E}[\cdots])|\psi\rangle\|_{\ell_2}^q)^{1/q} \leq 2\sqrt{(q-1)(t\lambda)^2/N}$. With the negligible bias (5) and Markov's inequality (optimizing q), for $N \geq (t\lambda)^2$,

$$\max_{\rho} \mathbb{E}[\tfrac{1}{2}\|\mathcal{U}(\rho) - \mathcal{V}^{(N)}(\rho)\|_1] \leq 4\sqrt{\frac{t^2\lambda^2}{N}}, \quad \max_{\rho} \Pr[\tfrac{1}{2}\|\mathcal{U}(\rho) - \mathcal{V}^{(N)}(\rho)\|_1 > \epsilon] \leq \exp(-\frac{\epsilon^2 N}{32e t^2 \lambda^2}), \quad (7)$$

which is Theorem 2. The absence of d here is precisely why the count is n -free: the dimension-free vector inequality removes the n that Theorem 1 pays to cover all 2^n inputs at once.

4 Intuition and Generalization

Why the n appears (and is necessary). For the diagonal commuting Hamiltonian $H = 2^{-n} \sum_{p \in \{0,1\}^n} \alpha_p Z_p$ ($\alpha_p \in \{\pm 1\}$), $e^{-iH}|b\rangle = e^{-iS(b)}|b\rangle$ with phase $S(b) = 2^{-n} \sum_p c_p(b)$, $c_p(b) \in \{\pm 1\}$. Sampling N terms gives the Monte Carlo estimate $\hat{S}(b) = \frac{1}{N} \sum_k c_{p_k}(b)$, which by Hoeffding concentrates around $S(b)$ with deviation $\sim 1/\sqrt{N}$. For *one* basis state, $N = 1/\epsilon^2$ suffices (matching Theorem 2); to control *all* 2^n states, a union bound multiplies the failure probability by 2^n , so $N = n/\epsilon^2$ is needed (matching Theorem 1). Thus $n = \log(2^n)$ is the log-count of states controlled at once, the same role $\log(2d)$ plays in Proposition 1. The dependence is tight at both extremes: single-site $H = \sum_k Z_k$ gives, via a multinomial central-limit argument, a matching lower bound $\mathbb{E}\|U - V_N \cdots V_1\| \gtrsim \sqrt{(n-1)(t\lambda)^2/N}$, while for the all-strings $H = \sum_p Z_p$ the diagonal entries become asymptotically independent (a Walsh–Hadamard/CLT analysis), making the union bound tight and again forcing $\sqrt{n}$. Thus commuting Hamiltonians saturate the upper bound, so the factor n is not a proof artifact.

Beyond qDRIFT. The proof used only two properties of the random factors V_k approximating $U = U_N \cdots U_1$: *causality* (each V_k depends on the past, not the future) and *accurate steps*, $\|V_k - \mathbb{E}_{k-1} V_k\| \leq a_k$, $\|\mathbb{E}_{k-1} V_k - U_k\| \leq b_k$. The same analysis then gives $\|\mathcal{U} - \mathcal{V}\|_{\diamond} \leq 2 \sum_k a_k$ (worst case), $\mathbb{E}\|\mathcal{U} - \mathcal{V}\|_{\diamond} \lesssim \sqrt{n \sum_k a_k^2} + 2 \sum_k b_k$ (typical), $\mathbb{E}\|\mathcal{U}(\rho) - \mathcal{V}(\rho)\|_1 \lesssim \sqrt{\sum_k a_k^2} + 2 \sum_k b_k$ (fixed input), and $\|\mathcal{U} - \mathbb{E}\mathcal{V}\|_{\diamond} \leq 2 \sum_k b_k$ (average). The bias is governed by $\sum_k b_k$, the fluctuation by $\sqrt{\sum_k a_k^2}$, with $\sqrt{n}$ only in the worst-case channel

norm. This covers randomly permuted Suzuki formulas and randomized compiling. A by-product is a sharpened mixing lemma, $\|\mathcal{U}_k - \mathbb{E}\mathcal{V}_k\|_\diamond \leq 4\|U_k - \mathbb{E}V_k\|$, removing the $\frac{1}{2}a^2$ term of the Hastings–Campbell version.

Two conceptual remarks. First, the $1/\epsilon$ vs. $1/\epsilon^2$ split mirrors the distinction between *weak* and *strong* error for stochastic differential equation solvers: the averaged channel tracks $\mathbb{E}[V_N \cdots V_1]$ (a weak, mean quantity, convergence order 1 in $1/N$), whereas a single realization must control $V_N \cdots V_1$ itself (a strong, pathwise quantity, order $1/2$). Second, the guarantees are *existential*: Theorem 1 shows that a good circuit is sampled with high probability, but certifying that a given drawn circuit is accurate is a separate, harder problem — a gap that motivates a posteriori estimators and derandomized (low-discrepancy) constructions. Because the count is quadratic in t , qDRIFT is most attractive for short-time simulation and for Hamiltonians with many terms, where deterministic formulas pay the explicit L .

5 Numerical Experiments

We now turn to the numerical part. The experiments below are meant to check the scaling picture from the theorems, not the hidden constants in the bounds. The point is simple: asking one random product formula to work for every input state is harder than asking it to work for one fixed input state. After normalizing the Hamiltonian strength λ , the fixed-input error should change only mildly with the system size.

5.1 Review of the theoretical prediction

The simulation problem starts from the Schrödinger equation

$$i \frac{d}{dt} |\psi(t)\rangle = H |\psi(t)\rangle,$$

whose solution for a time-independent Hamiltonian is

$$|\psi(t)\rangle = U(t) |\psi(0)\rangle, \quad U(t) = e^{-itH}.$$

Digital Hamiltonian simulation replaces this ideal unitary by a product of simpler gates. For a Hamiltonian $H = \sum_{j=1}^L h_j$, qDRIFT uses random sampling instead of cycling through all L terms in every layer. Let

$$\lambda = \sum_{j=1}^L \|h_j\|, \quad p_j = \frac{\|h_j\|}{\lambda}.$$

One sampled product formula is generated as Algorithm 1. This is the algorithmic content of the flow chart in the original paper.

Algorithm 1 qDRIFT random product formula

Require: Hamiltonian $H = \sum_{j=1}^L h_j$, evolution time t , number of steps N

- 1: Compute $\lambda = \sum_{j=1}^L \|h_j\|$ and probabilities $p_j = \|h_j\|/\lambda$
 - 2: Initialize $V \leftarrow I$
 - 3: **for** $k = 1, 2, \dots, N$ **do**
 - 4: Sample an index $j_k \in \{1, \dots, L\}$ with probability p_{j_k}
 - 5: Set $X_k = (\lambda/\|h_{j_k}\|)h_{j_k}$
 - 6: Apply $V_k = \exp(-i(t/N)X_k)$
 - 7: Update $V \leftarrow V_k V$
 - 8: **end for**
 - 9: **return** the random product formula $V = V_N \cdots V_1$
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The issue in the paper is whether qDRIFT works only after averaging over many such products, or whether a typical single draw already gives a useful approximation. The two main theorems reviewed above give different answers depending on the error criterion. In the all-input or worst-case setting, Theorem 1 states that a single sampled product formula satisfies, up to logarithmic and constant factors,

$$\epsilon_{\text{all}} \lesssim \sqrt{\frac{n\lambda^2 t^2}{N}},$$

where n is the number of qubits. The factor n comes from controlling all input states and observables simultaneously. In the fixed-input setting, Theorem 2 gives the dimension-free behavior

$$\epsilon_{\text{fixed}} \lesssim \sqrt{\frac{\lambda^2 t^2}{N}},$$

provided the input state is fixed independently of the sampled gates. Therefore, once t , λ , and N are held fixed, the theory predicts that the fixed-input curve should be approximately flat as the system size grows, while the all-input curve can grow with system size.

5.2 Error metrics

Each plotted point is the average over 50 independent random product formulas. The evolution time is fixed to $t = 2$. Two errors are measured. For the worst-case setting, the experiment computes the operator norm

$$\epsilon_{\text{all}} = \|U - V_N \cdots V_1\|_{\infty},$$

which controls all input states and all observables through Lemma 1. For the fixed-input setting, the input state is a tensor product of single-qubit Haar-random states, chosen independently of the qDRIFT gates, and the experiment computes

$$\epsilon_{\text{fixed}} = \|U|\psi\rangle - V_N \cdots V_1|\psi\rangle\|_{\ell_2}.$$

The relative-size plots normalize each curve by its smallest-system value. The dotted line in the all-input relative plot is an empirical linear fit to the measured data, not a theoretical equality; the dotted line in the fixed-input relative plot is the empirical mean, included to show whether the curve is approximately horizontal.

5.3 One-dimensional Heisenberg chain

The first experiment follows the numerical setting in the paper. The Hamiltonian is the normalized nearest-neighbor Heisenberg chain

$$H = \frac{1}{n-1} \sum_{i=1}^{n-1} (X_i X_{i+1} + Y_i Y_{i+1} + Z_i Z_{i+1}),$$

where $n = 4, 5, 6, 7, 8$ is the number of qubits. This normalization gives $\lambda = 3$ for all system sizes. The gate counts are $N = 20, 40, 80, 120, 160, 240, 320$, and the system-size comparison uses $N = 160$.

This is the model and normalization used for Figure 3 in Chen–Huang–Kueng–Tropp. The paper plots the error as N varies and also plots ϵ_n/ϵ_4 at $N = 160$, averaging over 50 independent random product formulas. We keep these choices. The only change is that the gate-count plots also show all five sizes $n = 4, 5, 6, 7, 8$, so the size dependence can be seen directly in the convergence curves. In the relative-error panels, I use empirical guide lines instead of copying the paper’s ideal scaling guide; for these small Heisenberg chains the worst-case upper bound is not saturated exactly.

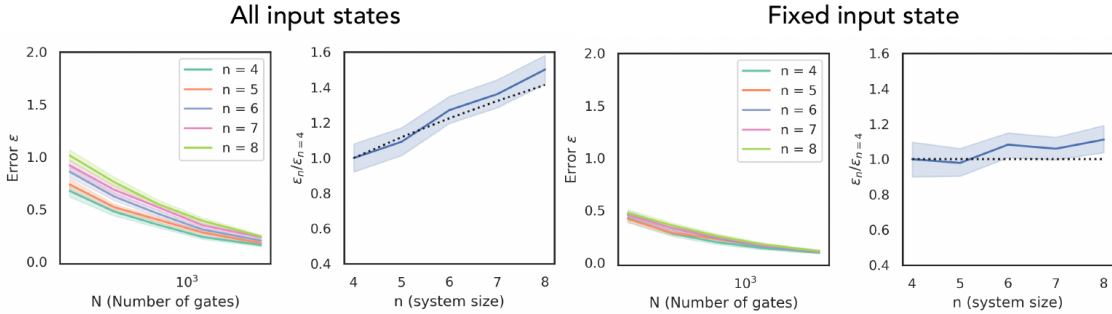


Figure 3. *Numerical experiments for simulating 1D Heisenberg model under different gate count N .* In *All input states* (left), we consider $\epsilon = \|U - V_N \dots V_1\|_\infty$, which considers the error over all input states and observables. In *Fixed input state* (right), we consider $\epsilon = \|U|\psi\rangle - V_N \dots V_1|\psi\rangle\|_{\ell_2}$, which considers the error over all observables. The input state $|\psi\rangle$ is chosen to be the tensor product of single-qubit Haar-random states. For both *All input state* and *Fixed input state*, we give an additional plot showing how the error ϵ increases as the system size n increases for a fixed number of gates $N = 160$. The y -axis is normalized using the average error for system size $n = 4$ over 50 independent runs. Bounds in Theorem 1 and 2 show that the relative error $\epsilon_n/\epsilon_{n=4}$ scales as $\sqrt{n}/4$ for *All input state* and stays as constant 1 for *Fixed input state*, which are shown as the dotted lines. The shaded regions are the standard deviation over 50 independent runs.

Figure 1: Original numerical experiment from Chen–Huang–Kueng–Tropp, cropped from Figure 3 of the paper. It uses the same normalized one-dimensional Heisenberg setting and compares all-input and fixed-input errors.

Figure 1 and Figure 2 tell the same story. The fixed-input curves move much less with n than the all-input curves. The differences are mostly presentational. The original figure highlights the $n = 4$ gate-count sweep and puts the size comparison in separate panels; my plot overlays $n = 4, 5, 6, 7, 8$ in the gate-count sweep. The paper’s dotted line is a theoretical scaling guide, while mine is a fit to the finite-size data. Since only 50 samples are used in both cases, the shaded standard-deviation bands are still visibly noisy.

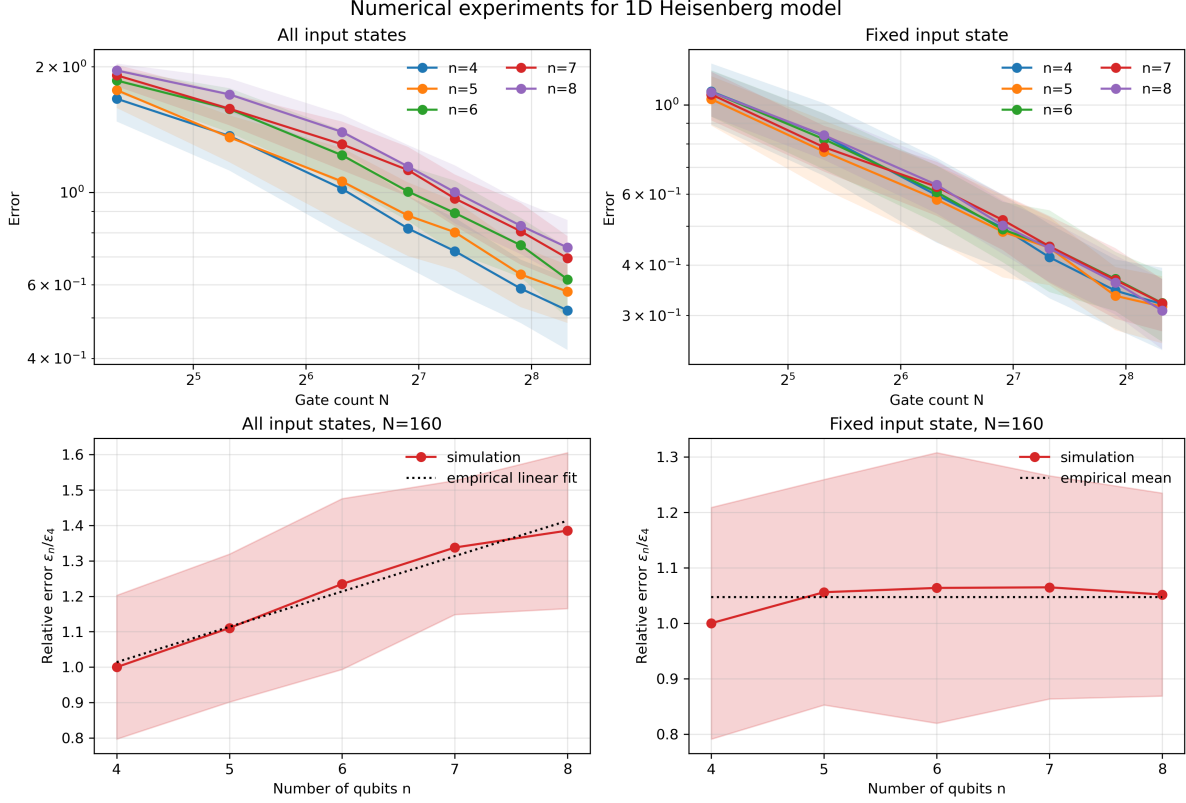


Figure 2: Our qDRIFT simulation of the normalized one-dimensional Heisenberg chain. The top row compares the decay of the error as the gate count N increases. The bottom row compares the normalized error as the number of qubits increases at fixed $N = 160$. The fixed-input error is much flatter than the all-input error, in agreement with the dimension-free fixed-input concentration bound.

The trend is the one predicted by the two theorems. At fixed N , the all-input error grows with the number of qubits, as suggested by the dimension factor in Theorem 1. The fixed-input curve is close to flat. Thus the factor n should be viewed as the price of controlling all inputs at once, not as a basic cost of the qDRIFT sampling step.

5.4 SYK model

The second experiment replaces the Heisenberg Hamiltonian by a Sachdev–Ye–Kitaev (SYK) Hamiltonian. For n Majorana fermions,

$$H = \sum_{1 \leq i < j < k < l \leq n} J_{ijkl} \chi_i \chi_j \chi_k \chi_l,$$

where the couplings are sampled from a centered Gaussian distribution with

$$\overline{J_{ijkl}^2} = \frac{3!J^2}{n^3}.$$

The Majorana operators are represented by the Jordan–Wigner construction

$$\chi_{2r+1} = \frac{1}{\sqrt{2}} Z_1 \cdots Z_r X_{r+1}, \quad \chi_{2r+2} = \frac{1}{\sqrt{2}} Z_1 \cdots Z_r Y_{r+1}.$$

Thus $n = 8, 10, 12, 14, 16$ Majorana modes correspond to 4, 5, 6, 7, 8 qubits. After drawing the Gaussian couplings, the Hamiltonian is rescaled so that $\lambda = 3$ for every system size. This extra normalization is important: without it, the qDRIFT strength λ grows with the number of SYK terms, and a fixed N comparison would mix system-size effects with a changing Hamiltonian strength.

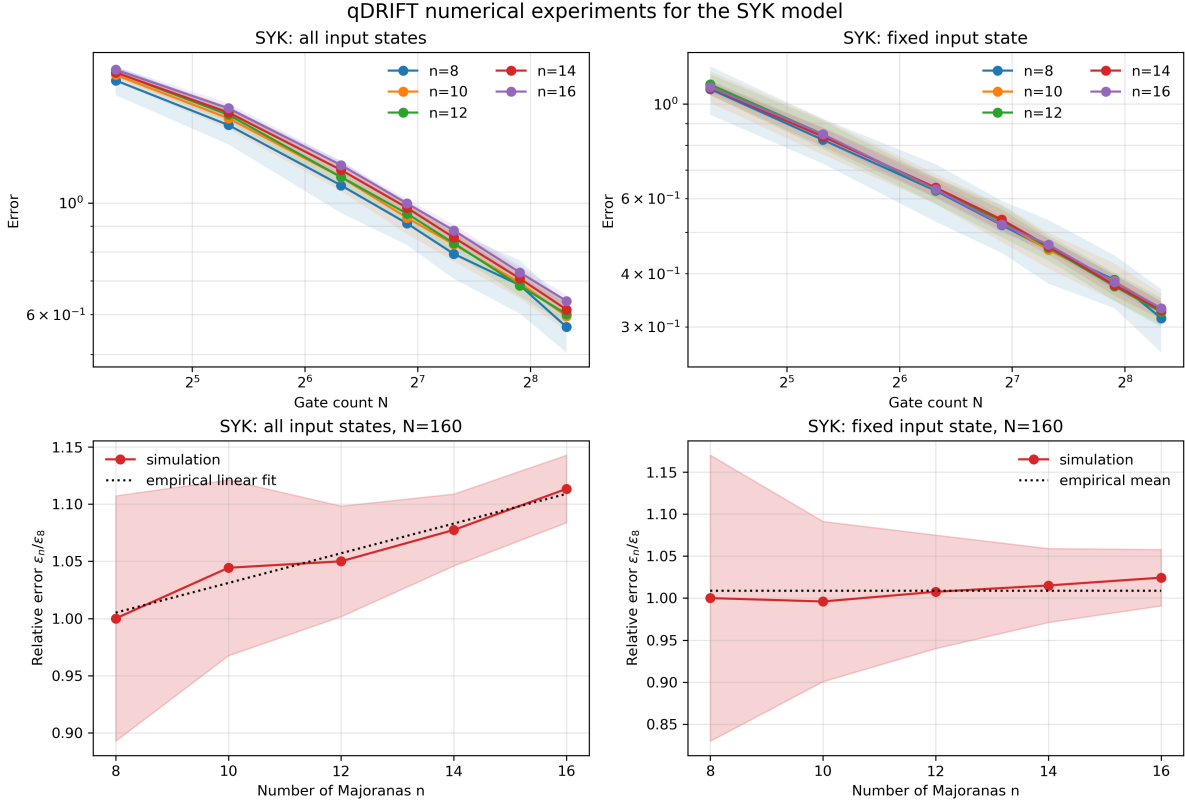


Figure 3: qDRIFT simulation of the SYK model after rescaling each instance to $\lambda = 3$. The same gate counts and error metrics as in the Heisenberg experiment are used. Once the Hamiltonian strength is fixed, the fixed-input error is again nearly independent of system size, while the all-input error shows a stronger size dependence.

The SYK test gives a useful check beyond a local spin chain. The Hamiltonian contains all four-Majorana interaction terms and becomes highly nonlocal after the Jordan–Wigner mapping. Still, after rescaling to the same λ , the fixed-input curve is nearly flat. This agrees with Theorem 2, where the bound depends on λ , t , and N , but not explicitly on the Hilbert-space dimension. The all-input curve keeps a clearer size dependence, which is what the matrix-martingale bound suggests.

6 Summary

Viewing QDRIFT as a random walk on $U(d)$, the error splits into a deterministic bias (controlled to $O(t^2\lambda^2/N)$ by matrix-exponential estimates and telescoping) and a random fluctuation (controlled to $O(\sqrt{n t^2\lambda^2/N})$ by a matrix martingale). Restricting to a fixed input replaces the matrix martingale by a dimension-free vector martingale, removing n and giving an n -fold reduction with no knowledge of the state. The L -independence of QDRIFT thus stems from random sampling itself; averaging only upgrades accuracy from $1/\epsilon^2$ to $1/\epsilon$. Section 5 checks the scalings $E_{\text{worst}} \sim \sqrt{n/N}$ and $E_{\text{fixed}} \sim N^{-1/2}$ numerically on Heisenberg and SYK Hamiltonians.

A Additional Comments

A.1 A correction to the causality condition

Section C of the original paper extends the concentration argument to general products of random unitaries. Its causality condition is written in the form

$$\Pr[V_k \mid V_N, \dots, V_{k+1}, V_{k-1}, \dots, V_1] = \Pr[V_k \mid V_{k-1}, \dots, V_1], \quad 1 \leq k \leq N.$$

Read literally as a conditional-independence statement, this condition is stronger than intended. In fact, if it holds for every k , then $V_1, V_2, \dots, V_N$ must be mutually independent.

The reason is just the chain rule, written in reverse order:

$$\Pr[V_1, \dots, V_N] = \prod_{k=1}^N \Pr[V_k \mid V_N, \dots, V_{k+1}].$$

Now apply the displayed condition with no past variables fixed. It gives

$$\Pr[V_k \mid V_N, \dots, V_{k+1}] = \Pr[V_k].$$

Therefore

$$\Pr[V_1, \dots, V_N] = \prod_{k=1}^N \Pr[V_k],$$

which is exactly mutual independence. This is not the intended adapted-random-process setting, because in an adaptive product formula V_k is allowed to depend on the previously sampled gates.

A more accurate way to state the adapted setting is to separate the past from the new randomness. Let $U_1, U_2, \dots, U_N$ be mutually independent random variables, and set

$$V_k = f_k(V_1, V_2, \dots, V_{k-1}, U_k), \quad 1 \leq k \leq N,$$

where each f_k is a deterministic measurable function. Equivalently, with the filtration

$$\mathcal{F}_k = \sigma(U_1, \dots, U_k) = \sigma(V_1, \dots, V_k, U_1, \dots, U_k),$$

the random unitary V_k is $\mathcal{F}_k$ -measurable, and the only new randomness entering at time k is U_k , which is independent of $\mathcal{F}_{k-1}$. This is the usual causality or adaptedness condition needed for martingale arguments: the present choice may depend on the past, but it cannot depend on future random choices.

A.2 Group contribution

Yuan Lai wrote the theory part. Tang Jinqi handled the numerical experiments and code.

A.3 AI usage report

AI was used in three places: reading and organizing the paper, debugging the experiment code, and polishing the English. The authors checked the mathematical statements, the experimental setup, and the final conclusions.

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